Process or Materials Review Request Form

Request Title: Use and Storage of Copper Electroplating Solution in SNF

Requestor: Joanna Nassar **Date:** 4/08/2019

Non-SNF staff, please complete the following:

Badger ID: joannamn

Company name/PI: Prof. Barbara Block

Send completed form to: snf-promcommittee@lists.stanford.edu

1. **The chemical or material:**

High-speed bright Copper Electroplating solution. CAS No: 7758-99-8 and CAS No: 7664- 93-9.
Storage Group G: Not Intrinsically Reactive or Flammable or Combustible
Health: 3 Flammability: 0 Reactivity: 0 Personal Protection: C
(attached SDS in email).

2. **Vendor/manufacturer info:**

Nano3D Systems (Electrochemical Deposition); URL: https://nano3dsystems.com/product/high-speed-bright-copper-electroplating-solution/; Adress: Nano3D Systems, LLC, 1110 NE Circle Blvd. Bldg. 11 Corvallis, OR 97330; phone: (503) 927-4766; info@nano3dsystems.com

3. Reason for request: to be used for multi-level system interconnects in flexible electronics. Other metals and deposition options were previously tested, but electroplated copper showed to be best for desired low stress, high quality and high conductivity. Experimental procedure previously used in an industrial electroplating bench in a cleanroom, using the following conditions: 0.08 A current and 15 mins for 4 um thickness. Current desired thickness is 5 um.

Process Flow: Will be used in the fume hood of PDMS room 155A (discussed with Michelle)
Plating solution will be dispensed into a 2000 mL pyrex breaker
Solution will be filled to the 1500 mL mark
Beaker will be placed onto a magnetic stirrer available in room 155A in the fumehood.
Power supply: (30 V max; 5 A max) ? my usage would be up to 1A of current max. https://www.amazon.com/gp/product/B00ZBCLJSY/
f=ox_sc_act_title_11?smid=AMH4W1K8OCGMX&psc=1
2 copper anodes will be used
4 inch wafer will be suspended in the beaker with the plating solution using a rod and clamps

- 24 gauge copper wires will be used for connections to the main lead connections for the power supply
- Room temperature process only. No heating will be required.
- Another beaker will be filled with DI water and will be used for cleaning the wafer from the plating solution. Then drying with N2.

Appropriate protective gear like gloves and apron will be used at all times.
Next steps on my wafer will all be processed in ALL and CONT categories in the cleanroom.

5. Amount and form: 1 gallon poly bottle, meeting UN requirements. Liquid Form.

6. Storage: Storage Group: G: Not Intrinsically Reactive or Flammable or Combustible; Need to store in SNF on top of one of the Pass-through Carts. Store in a cool, dry area. Use with adequate ventilation. Keep container tightly closed when not in use.

7. Disposal:
Spill, Leak or Release: Isolate spill area and avoid tracking through liquid. Dike and prevent runoff to drains or sewers. For small spills, cover with lime or soda ash and then scoop into polyethylene drums for later disposal.
Large spill may be pumped directly into a storage container for later disposal. Do not wash residue to drain or sewer. Copper sulfate solution may deteriorate concrete.

After plating process, plating solution will be dispensed into a "waste" container that will be labeled by my name and with a tag of the solution for the material info. This solution will be reused over and over until no longer possible. Then it will be taken to the waste collection area of other hazardous solutions in SNF for final disposal.

Gloves and wipes contaminated with the plating solution will be placed in a plastic Ziploc and thrown in the hazardous waste bin. For small spills in the fumehood, absorb on cleanroom wipes and dispose in a ziploc and hazardous waste bin.

- Cont Teflon tweezers will be used for this process.

Please refer to the SNF Lab members Safety Manual for the different methods of waste disposal that are available in the lab.

To be completed by the PROM COMMITTEE:

Special handling requirements:

On-going approval or staff interaction requirements:

Approved By/Date: